

Homework 4

EE 999 — Introduction to Control Systems

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Due Wednesday, October 7, 2026

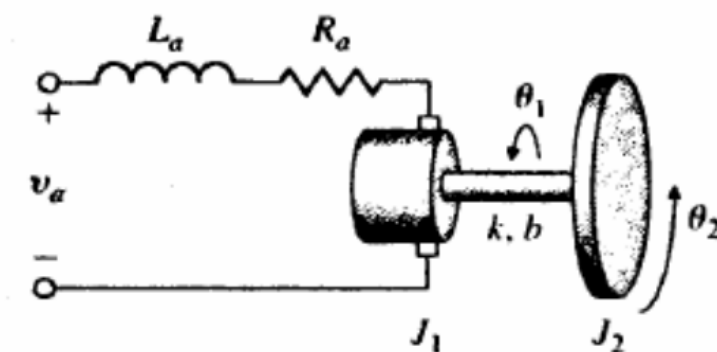
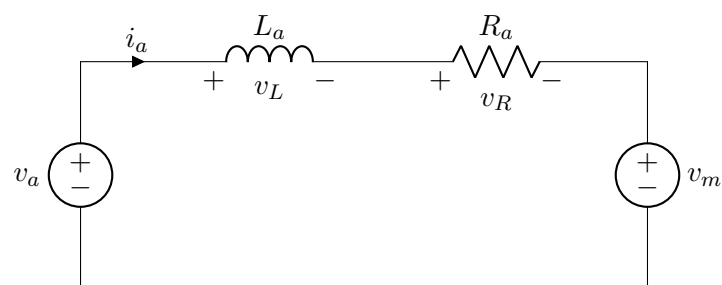
Problem A.1 — Equations of Motion
Figure 1: DC motor driving a load inertia J_2 

Figure 2: Electrical Circuit Model

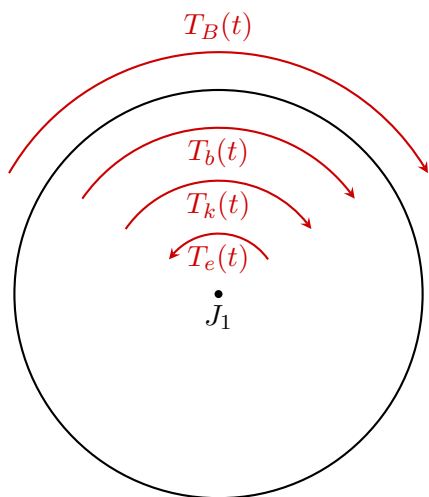


Figure 3: Free Body Diagram of Rotor

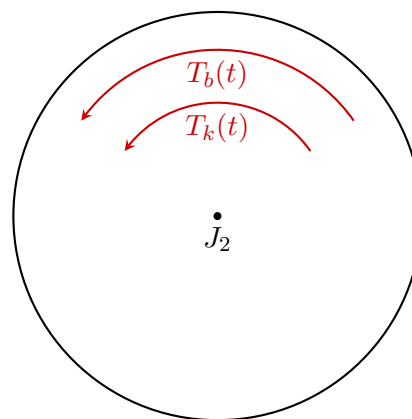


Figure 4: Free Body Diagram of Load

Electrical side:

$$v_a(t) = v_L(t) + v_R(t) + v_m(t)$$

$$v_L(t) = L_a \dot{i}_a(t)$$

$$v_R(t) = R_a i_a(t)$$

$$v_m(t) = K_e \dot{\theta}_1(t)$$

Rotor:

$$J_1 \ddot{\theta}_1(t) = T_{net.J_1}(t) = T_e(t) - T_B(t) - T_k(t) - T_b(t)$$

$$T_e(t) = K_t i_a(t)$$

$$T_B(t) = B \dot{\theta}_1(t)$$

$$T_k(t) = k(\theta_1 - \theta_2)$$

$$T_b(t) = b(\dot{\theta}_1 - \dot{\theta}_2)$$

Load:

$$J_2 \ddot{\theta}_2(t) = T_{net.J_2}(t) = T_b(t) + T_k(t)$$

Standard Forms

$$v_a(t) = L_a \dot{i}_a(t) + R_a i_a(t) + K_e \dot{\theta}_1(t)$$

$$J_1 \ddot{\theta}_1(t) = K_t i_a(t) - B \dot{\theta}_1(t) - k(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2)$$

$$J_1 \ddot{\theta}_1(t) + B \dot{\theta}_1(t) + b \dot{\theta}_1 + k \theta_1(t) = K_t i_a(t) + k \theta_2 + b \dot{\theta}_2$$

$$J_2 \ddot{\theta}_2(t) = k(\theta_1(t) - \theta_2(t)) + b(\dot{\theta}_1(t) - \dot{\theta}_2(t))$$

$$J_2 \ddot{\theta}_2(t) + b \dot{\theta}_2(t) + k \theta_2(t) = k \theta_1(t) + b \dot{\theta}_1(t)$$

Problem A.2 — Electrical Circuit Model

Mechanical	Units	Electrical	Units
$\dot{\theta}(t) = v(t)$	rad/s	Voltage	V
$T(t) = i(t)$	N·m	Current	A
$J = C$	kg·m ²	Capacitance	F
$B = 1/R$	N·m·s/rad	Conductance	S
$k = 1/L$	N·m/rad	Inverse inductance	1/H

Table 1: Firestone analogy: signal and unit correspondence.

Mechanical	Electrical
$T = J\ddot{\theta}$	$i = C\dot{v}$
$T = B\dot{\theta}$	$i = v/R$
$T = k\theta = k \int \dot{\theta} dt$	$i = \frac{1}{L} \int v dt$

Table 2: Element laws under the Firestone analogy.

Current equation for the first inertia:

$$\begin{aligned}
 J_1 \ddot{\theta}_1(t) &= K_t i_a(t) - B \dot{\theta}_1(t) - k(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2) \\
 k_t i_a(t) &= J_1 \ddot{\theta}_1(t) + B \dot{\theta}_1(t) + k(\theta_1 - \theta_2) + b(\dot{\theta}_1 - \dot{\theta}_2) \\
 I_e(t) &= C_1 \dot{v}_1(t) + \frac{1}{R_B} v_1(t) + \frac{1}{L_k} \left(\int v_1(t) - \int v_2(t) \right) + \frac{1}{R_b} (v_1(t) - v_2(t))
 \end{aligned}$$

and for the second:

$$\begin{aligned}
 J_2 \ddot{\theta}_2(t) &= k(\theta_1(t) - \theta_2(t)) + b(\dot{\theta}_1(t) - \dot{\theta}_2(t)) \\
 J_2 \ddot{\theta}_2(t) &= \frac{1}{L_k} \left(\int v_1(t) - \int v_2(t) \right) + \frac{1}{R_b} (v_1(t) - v_2(t))
 \end{aligned}$$

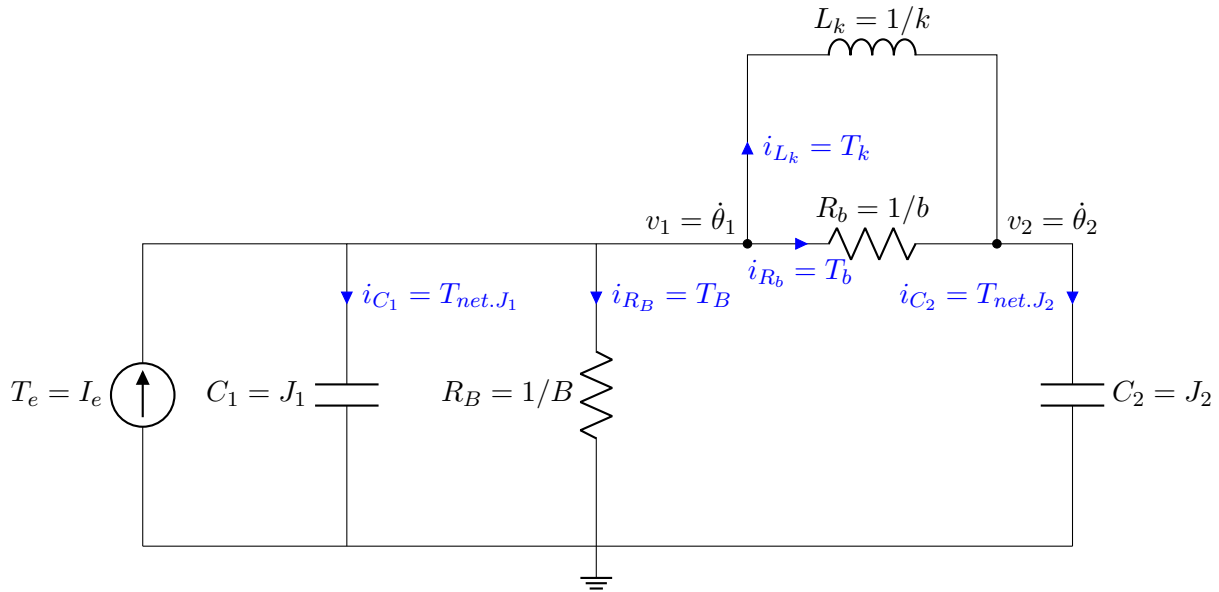


Figure 5: Firestone analogous circuit, with branch currents

Problem B — Units Balancing

$$J_m \ddot{\theta}_m + b \dot{\theta}_m = K_t i_a$$

$$(kg \cdot m^2) \left(\frac{rad}{s^2} \right) + \left(\frac{N \cdot m \cdot s}{rad} \right) \left(\frac{rad}{s} \right) = \left(\frac{N \cdot m}{A} \right) (A)$$

$$(N \cdot m \cdot \cancel{s}) \left(\frac{rad}{\cancel{s^2}} \right) + \left(\frac{N \cdot m \cdot \cancel{s}}{rad} \right) \left(\frac{rad}{\cancel{s}} \right) = \left(\frac{N \cdot m}{A} \right) (\cancel{A}) = N \cdot m$$

Noting that K_t is the motor torque constant with units of $\frac{N \cdot m}{A}$, so the amps cancel, and radians are eliminated as being unitless.

Problem C.1 — Matlab

Code: hw_04_C1.m

```

1 x = linspace(0, 90, 1000);
2 q = 20*(exp(x/15) - 1) ./ (exp(x/15) + 1);
3
4 figure;
5 plot(x, q);
6 grid on;
7 xlabel("Valve angle in degrees");
8 ylabel("Flow rate in [m^3/s]");
9 title("Flow rate vs valve angle")

```

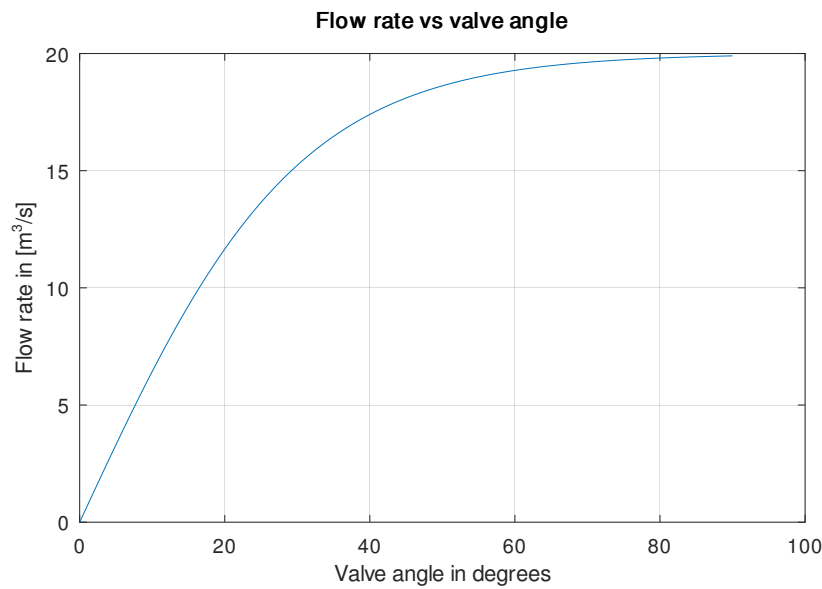


Figure 6: Plot of $q(t) = 20 \frac{\exp^{x(t)/15} - 1}{\exp^{x(t)/15} + 1}$

Problem C.2 — Linearization

Equation given in problem:

$$q(t) = 20 \frac{e^{x(t)/15} - 1}{e^{x(t)/15} + 1} \quad (1)$$

Linearization equations:

$$q(t) = f(x(t)) \quad (2)$$

$$x(t) = x_0 + \delta x(t) \quad (3)$$

$$q(t) = q_0 + \delta q(t) \quad (4)$$

$$\delta q(t) \simeq K_{xq} \delta x(t) \quad (5)$$

$$q(t) \simeq q_0 + K_{xq} \delta x(t) \quad (6)$$

$$K_{xq} = \dot{q}(t) \quad (7)$$

To find K_{xq} , take the derivative of q with respect to x :

$$\begin{aligned} u(t) &= e^{x(t)/15} - 1 \\ u'(t) &= \frac{1}{15} e^{x(t)/15} \end{aligned} \quad (8)$$

$$\begin{aligned} v(t) &= e^{x(t)/15} + 1 \\ v'(t) &= \frac{1}{15} e^{x(t)/15} \end{aligned} \quad (9)$$

$$q(t) = 20 \frac{u(t)}{v(t)} \quad (10)$$

$$\begin{aligned} K_{xq} &= 20 \frac{v\dot{u} - u\dot{v}}{v^2} \\ K_{xq} &= 20 \frac{\frac{1}{15} e^{x/15} (e^{x/15} + 1) - \frac{1}{15} e^{x/15} (e^{x/15} - 1)}{(e^{x/15} + 1)^2} \\ K_{xq} &= \frac{4}{3} \frac{2e^{x/15}}{(e^{x/15} + 1)^2} \end{aligned}$$

Code: hw_C_2.m

```

1 X = [20, 60];
2
3 for x = X
4     x_0 = x
5     q_0 = 20*(exp(x/15)-1)/(exp(x/15)+1)
6     num = 4*2*exp(x/15);
7     den = 3*(exp(x/15)+1)^2;
8     K_xq = num/den
9     disp(newline());
10 end

```

Output: output/hw_C_2.txt

```

x_0 = 20
q_0 = 11.656
K_xq = 0.4402

x_0 = 60
q_0 = 19.281
K_xq = 0.047101

```

Problem C.2.a — 20 degrees

$$q_{20}(t) = 11.656 + .4402 \cdot \delta x_{20}(t)$$

Problem C.2.b — 60 degrees

$$q_{60}(t) = 19.281 + .0471 \cdot \delta x_{60}(t)$$

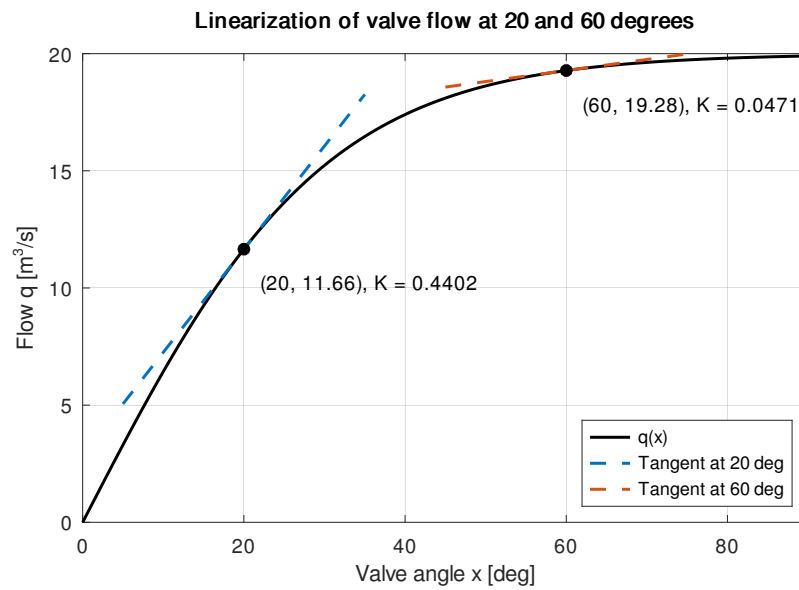
Problem C.2 appendix — Verification

Figure 7: Valve flow with the tangent lines (linearizations) at $x_0 = 20^\circ$ and $x_0 = 60^\circ$